HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Configured and integrated **SCADA systems** and **manufacturing execution systems** to align with ISA-95 stack requirements for client pilot lines.
- Developed **Python data pipelines** for processing and transforming sensor data from industrial equipment, enabling real-time operational insights.
- Implemented **REST APIs** using Java Spring Boot to facilitate data exchange between cloud platforms and on-premise industrial control systems.
- Designed and executed test plans for digital factory solutions, ensuring seamless integration and validation across **PLCs**, **edge devices**, and cloud environments.
- Managed **Git repositories** and conducted code reviews for a team of 3 engineers, adhering to collaborative development practices and maintaining code quality.
- Supported the deployment and troubleshooting of industrial networks, configuring **OPC UA** and **MQTT** protocols for reliable data transmission.
- Containerized backend services using **Docker** and deployed them to **AWS EC2** instances, managing application lifecycle and resource allocation.

SKILLS

Core Technologies – Python, Java, Spring Boot, SQL, REST APIs, Linux, Git, Docker, Jenkins

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Kubernetes

Databases – PostgreSQL, MySQL, Oracle, Time-series Databases

Industrial Automation – SCADA, PLC, MES, ISA-95, OPC UA, MQTT, Modbus, Profinet, EtherNet/IP

Data & Analytics – Power BI, Tableau, DAX, Power Query, pandas, NumPy, R, scikit-learn

DevOps & Tools – Jira, Confluence, Splunk, Dynatrace, CloudWatch

PROJECTS

Industrial IoT Data Pipeline & Dashboard

- Engineered a **Python-based data pipeline** to ingest and process telemetry data from a fleet of 50+ industrial sensors across multiple manufacturing sites.
- Developed a **Power BI dashboard** to visualize real-time operational metrics, including machine uptime, error rates, and production throughput, for stakeholder review.
- Integrated the data pipeline with a **PostgreSQL** database, structuring time-series data for efficient querying and historical analysis.

Manufacturing Systems Integration API

- Built a **Java Spring Boot microservice** to expose an API layer for a legacy MES system, enabling integration with newer shop-floor data collection tools.
- Implemented authentication and authorization mechanisms for the API, ensuring secure access for authorized client applications.
- Containerized the microservice using **Docker** and deployed it on an **AWS EC2** instance, establishing CI/CD pipelines with Jenkins for automated builds and deployments.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)